RAG ON DATABASE USING SLM (POC)

AI + Database

Prepared by Cross Platform Group - XPG

SLM (Small Language Model)

SLMs aren't just smaller models—they're the future of AI, bringing smart, private intelligence right to your device and daily life

September 22, 2025

EviCore
By EVERNORTH

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Why Small Language Model ?

+ **Portable & Accessible**: Runs offline, anywhere — even on 4GB RAM.
+ **Personalized**: Tailored for specific tasks — minimal, efficient.
+ **Cost-Effective**: No GPU needed, ideal for on-prem deployment too.
+ **Private & Secure**: Perfect for sensitive domains like healthcare.
+ **Sustainable**: Low energy use, eco-friendly alternative to large models.

List of some Small Language Model

SLM Name	Provider	RAM Requirement	Vector Dimensions / Parameters
Phi-1	Microsoft	~4GB	~1.3B parameters
Phi-3 Mini	Microsoft	~4GB	~3.8B parameters
Phi-4 Mini	Microsoft	~6–12GB	Mixture-of-LoRAs, multimodal
text-embedding-ada-002	OpenAI	Cloud-only	1536 dimensions
nomic-embed-text	Ollama	~4GB	768 dimensions
mxbai-embed-large	Ollama	~8–16GB	1024 dimensions
snowflake-arctic-embed	Ollama	~8–16GB	1024 dimensions
all-MiniLM-L6-v2	Microsoft (via Hugging Face)	~4GB	384 dimensions

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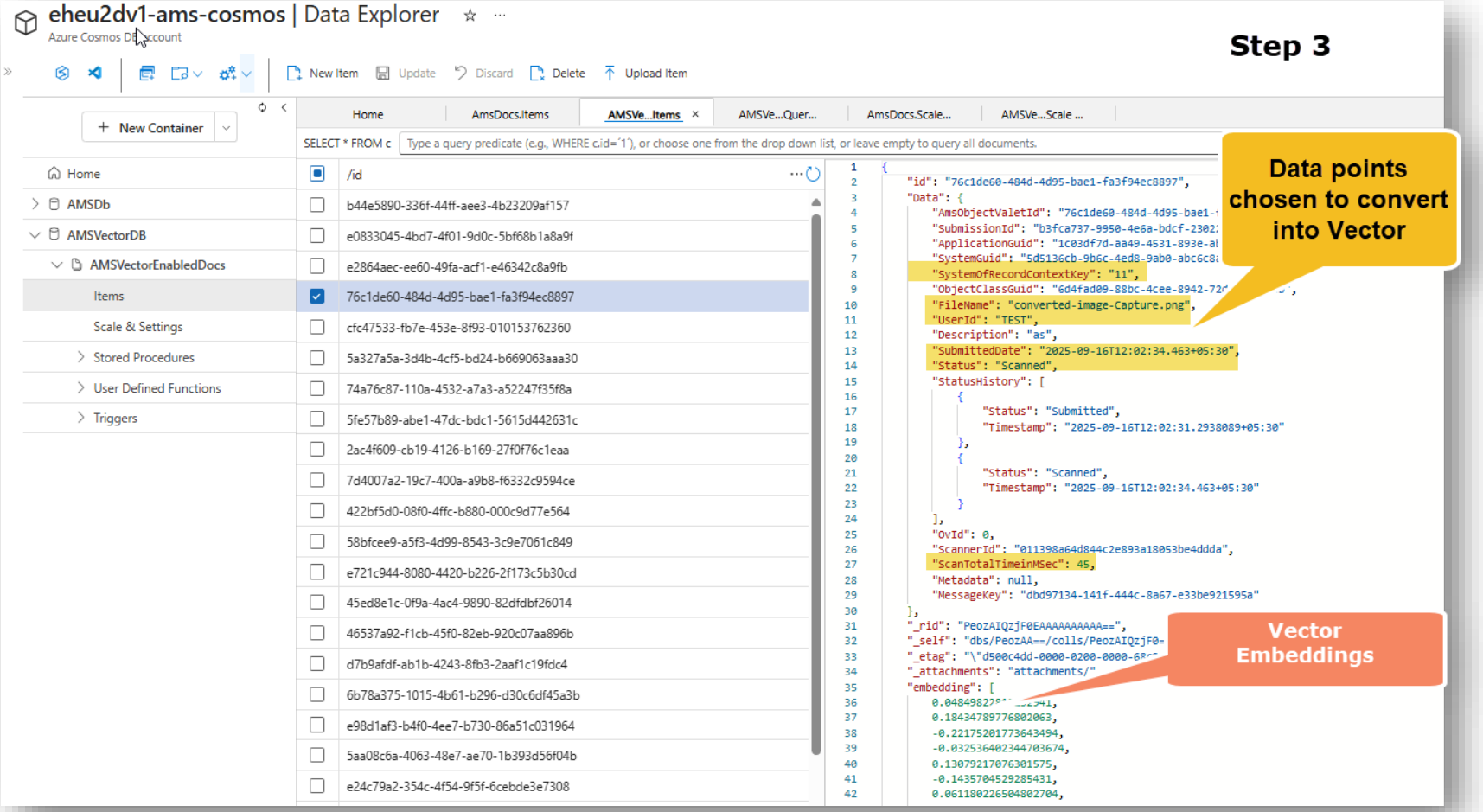
POC: RAG using SLM on database (No-SQL)

Embeddings

Why : Enables Large Language Models (LLMs) to understand and interact with our data using these representations

What : A mathematical representation of text, stored in a database

How : LLM's/SLM's generate those for us for a chosen data point via its endpoints



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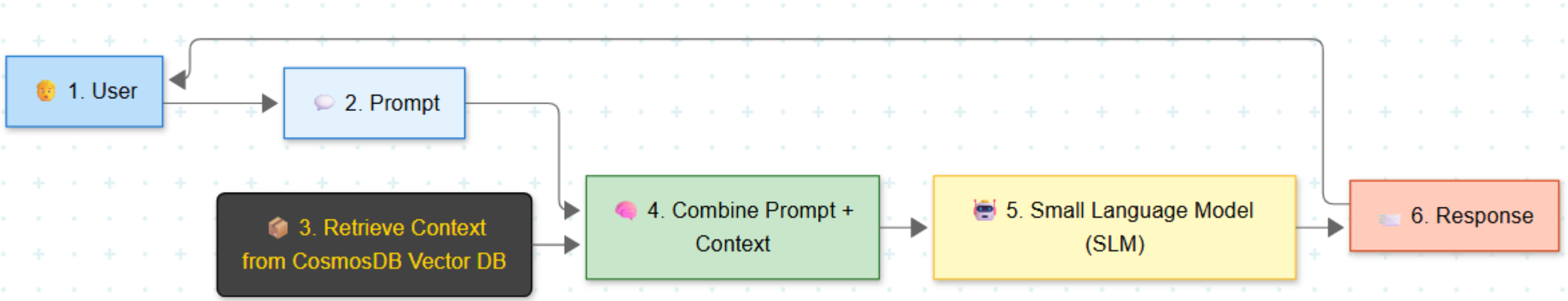
POC: RAG using SLM on database (No-SQL/SQL)

- Choosing the **right embedding dimension** for your database SQL / NO-SQL matters
  - 384–512: Good for mobile and fast inference
  - 768–1024: Balanced for most NLP tasks
  - >1024: High precision tasks like legal, medical, or multilingual embeddings

  - DEMO with RAG on Cosmos DB using SLM



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POC: RAG using SLM on database (No-SQL/SQL)

Integrate Natural Language Processing (NLP) on Cosmos DB / SQL Server data **without exposing data to external LLMs**

+ **Key Benefits:**

+ 🔍 **Contextual AI Experience**
  Enables deeper understanding across EviCore applications for enhanced NLP capabilities
+ 🔐 **Data Privacy & Compliance**
  Ensures sensitive data remains secure and compliant with internal EviCore policies
+ ⚡ **Accelerated Semantic Search**
  Improves search relevance and speed across datasets
+ 💰 **Cost Optimization**
  Reduces Cosmos DB Request Units (RU) consumption, leading to significant operational savings
+ 🚀 **Performance Efficiency**
  Delivers improved system performance with minimal overhead

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Thank you

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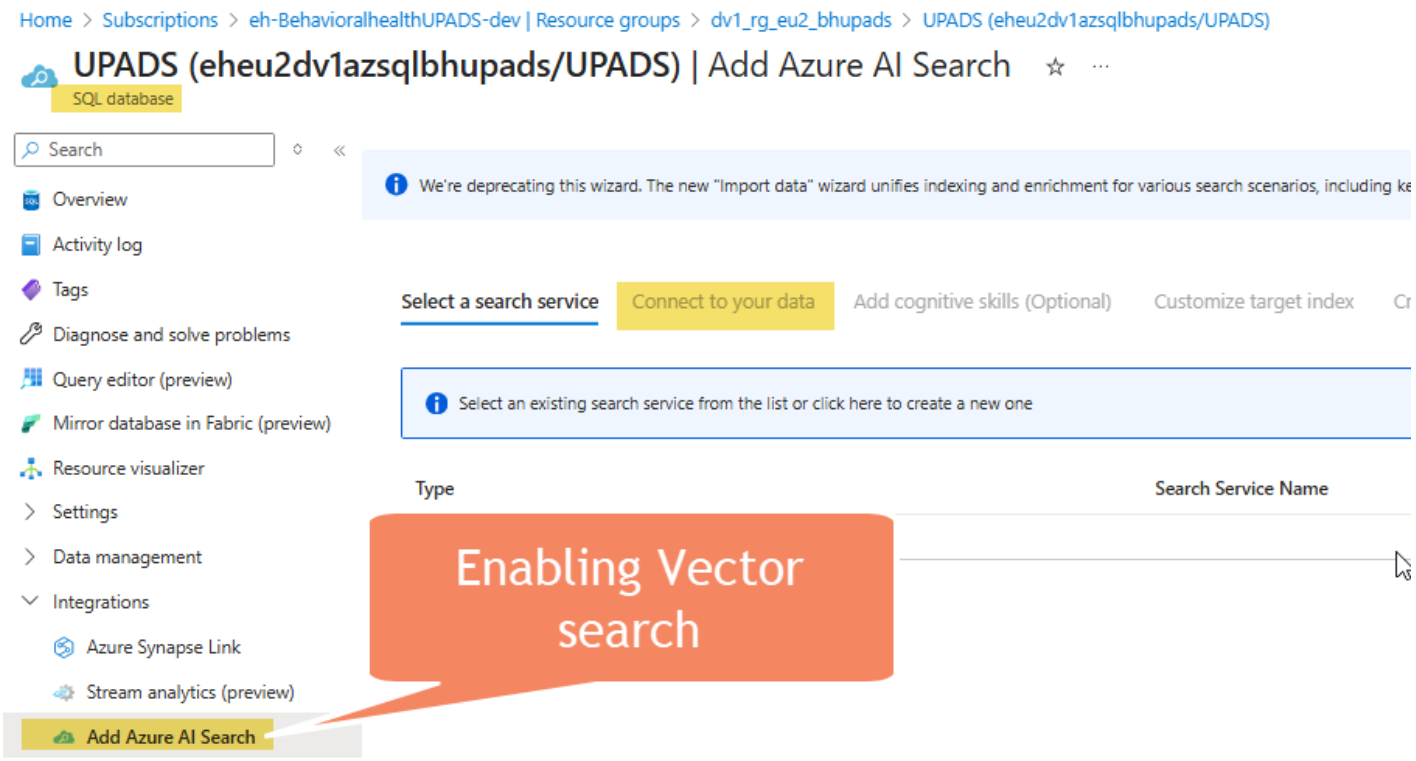
POC: RAG using SLM on database (SQL)

+ How to enable NLP capabilities search in Azure SQL DB Server

Enable vector search in Azure SQL Server DB & save embedding in separate tables in varbinary (8000) column



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